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Theory of Comparative Cost Advantage

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Trade Based on Comparative Cost Advantage: David Ricardo

- **Law of Comparative Advantage**
 - Even if one nation is less efficient than (has absolute disadvantage with respect to) the other nation in production of both commodities, there is still a basis for mutually beneficial trade.



Trade Based on Comparative Cost Advantage: David Ricardo

	U.S.	India
Wheat (bushels/labor hour)	6	1
Cloth (yards/labor hour)	4	2

- India has absolute disadvantage in both goods.
- Since India labor is half as productive in cloth but six times less productive in wheat compared to U.S., the India has a comparative advantage in cloth.
- U.S. has comparative advantage in wheat.
- $4C < 6W < 12C$

Comparative Advantage and Opportunity Costs

- The original idea of comparative advantage was based on the **labor theory of value**:
 - The value or price of a commodity depends exclusively on the amount of labor used to produce it.
- Can use the **opportunity cost theory** to explain comparative advantage:
 - The cost of a commodity is the amount of a second commodity that must be given up to release just enough resources to produce one additional unit of the first commodity.



Comparative Advantage and Opportunity Costs

- **Production Possibilities Frontier**

- A curve that shows alternative combinations of the two commodities a nation can produce by fully using all resources with best available technology.
- **Constant opportunity costs** arise when:
 1. Resources are either perfect substitutes for each other or used in fixed proportion in production of both commodities, and
 2. All units of the same factor are homogeneous.



Production possibility schedule

United States		United Kingdom	
wheat	cloth	wheat	cloth
180	0	60	0
150	20	50	20
120	40	40	40
90	60	30	60
60	80	20	80
30	100	10	100
0	120	0	120

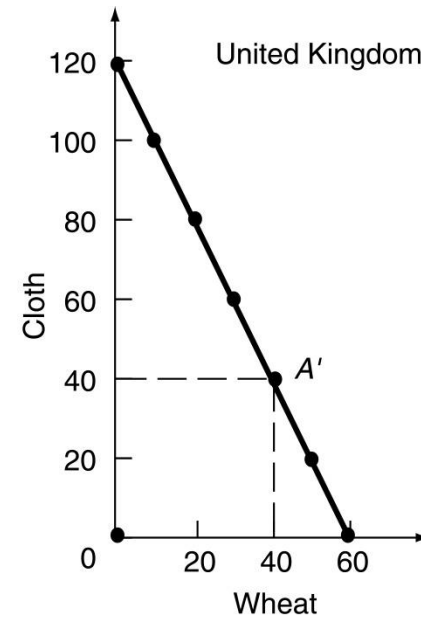
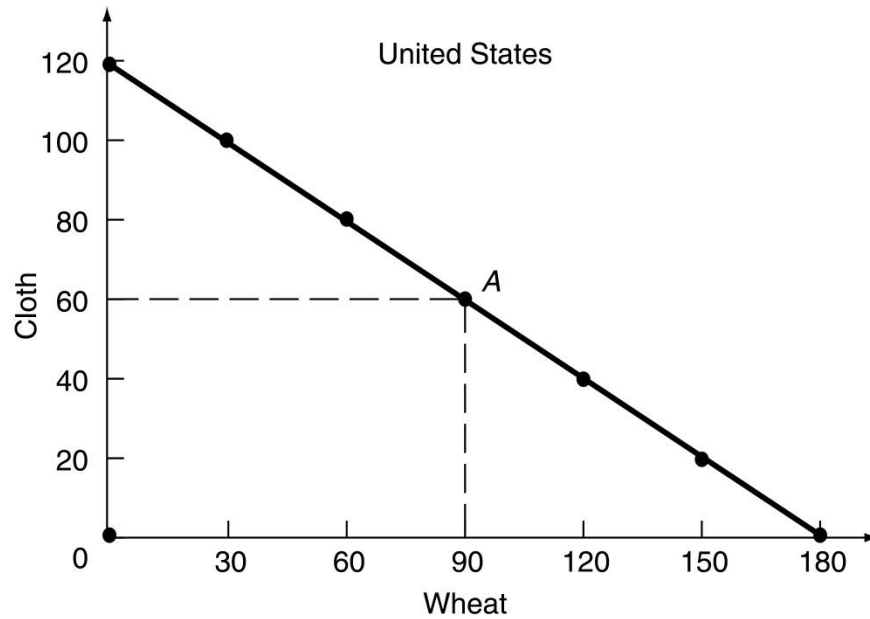


FIGURE 2-1 The Production Possibility Frontiers of the United States and the United Kingdom.

The Basis for and the Gains from Trade under Constant Costs

- In the absence of trade, a nation's production possibilities frontier also represents its *consumption frontier*.
- Increased output resulting from specialization and trade represents nations' gains from trade, allowing nations to consume outside production possibilities frontier.



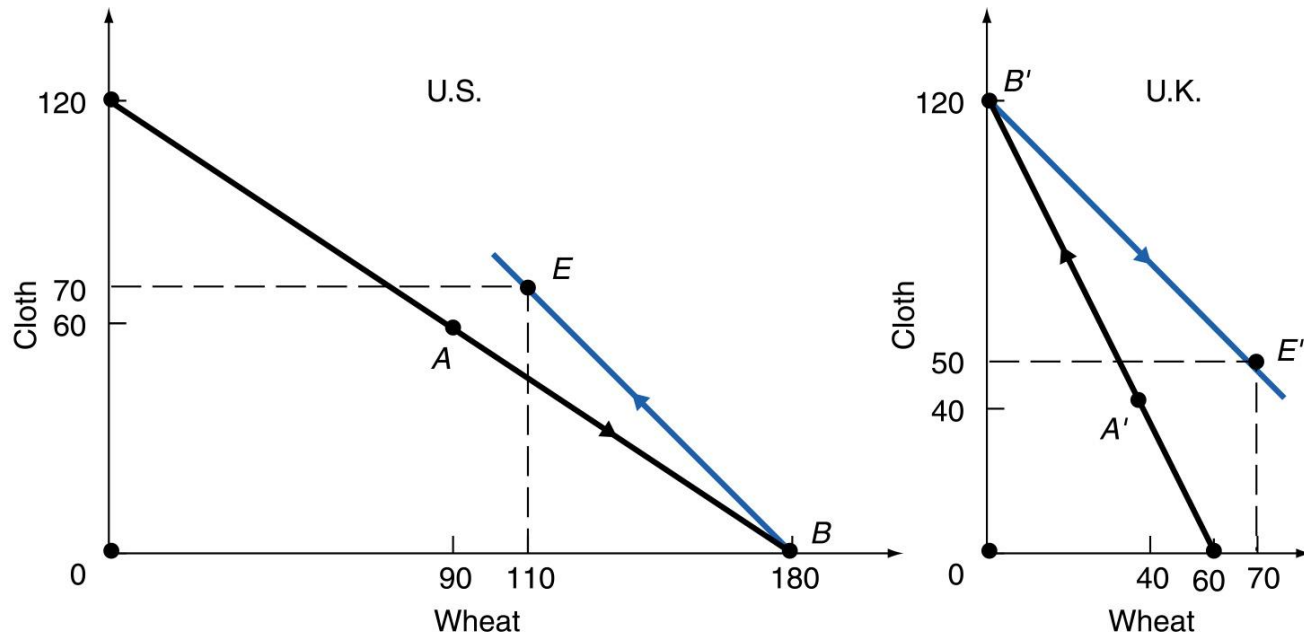
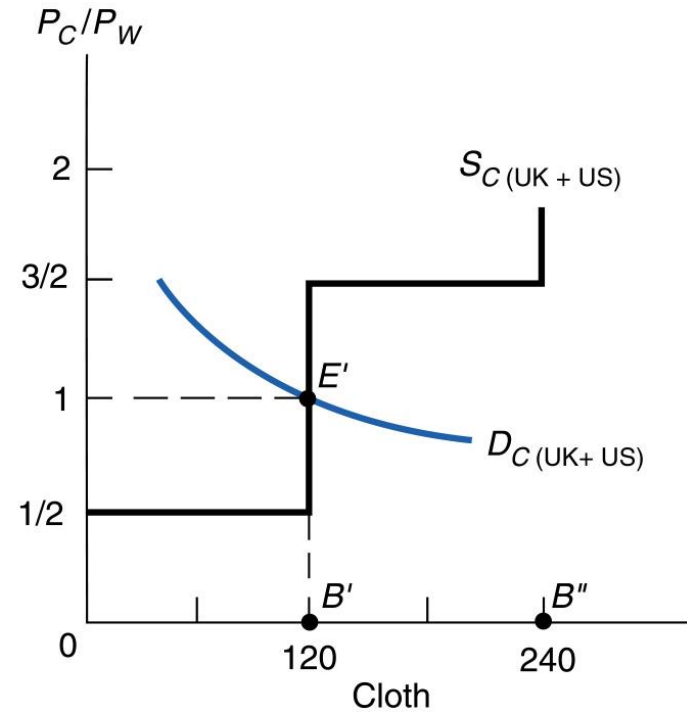
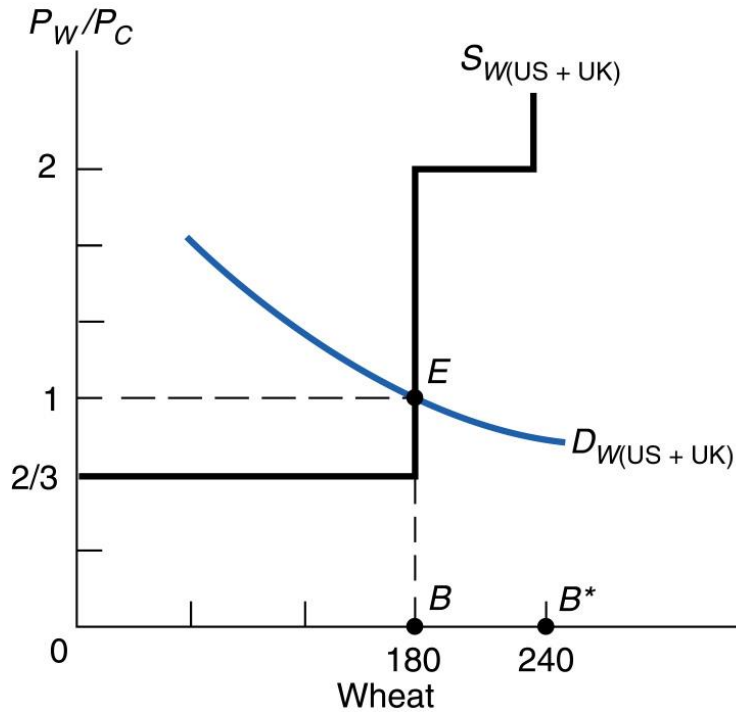


FIGURE 2-2 The Gains from Trade.

The Basis for and the Gains from Trade under Constant Costs

- Under constant cost conditions, nations will *completely specialize* in their comparative advantage .
- With complete specialization in both nations, the equilibrium-relative commodity price of each commodity lies between the pre-trade relative commodity price in each nation.





Equilibrium-Relative Commodity Prices with Demand and Supply.

- Revealed comparative advantage
- <https://www.financialexpress.com/india-news/india-must-focus-on-areas-it-has-comparative-advantage-rather-than-fear-rise-of-china/478594/>
- Balassa's revealed comparative advantage (RCA) index is useful.
- <https://doi.org/10.58537/jorangrau.2024.52.1.14> (Trends in Millets production, consumption and export from India) (Singh & Singh ,2024)



https://wits.worldbank.org/wits/wits/witshelp/Content/Utilities/e1.trade_indicators.htm

Revealed Comparative Advantage

- The **Revealed Comparative Advantage** is defined as the ratio of two shares.
- The numerator is the share of a country's total exports of the commodity of interest in its total exports, and the denominator is share of world exports of the same commodity in total world exports.
- The RCA takes a value **between 0 and (infinity)**. A Country is said to have a **revealed comparative advantage** if the value is **>1**.

- It is an index based on the Ricardian **comparative advantage** concept for calculating the relative competitiveness of a certain country in a certain class of goods or services using their trade flows as evidence.
- An index to indicate the **advantage** or **disadvantage** of a particular country in particular commodities or groups of commodities constructed with the help of trade flows.
- The concept of revealed comparative advantage (**Balassa 1965, 1977, 1979, 1986**) pertains to the relative trade performance of individual countries in particular commodities.

- **$RCA_{ij} = (X_{ij}/X_{wj}) / (X_i/X_w)$**
 - Where X_{ij} = ith country's export of commodity j
 - X_{wj} = world exports of commodity j
 - X_i = total exports of country i
 - X_w = total world exports

https://wits.worldbank.org/wits/wits/witshelp/Content/Utilities/e1.trade_indicators.htm

- the annual exports and imports data of 10 disaggregated service items from **2000 to 2013**, this paper (**Nath & Goswami 2018**) computes and analyzes various comparative advantage (CA) measures.
- The analysis reveals that India has had a CA in computer and information services and other business services (that include a wide range of information-intensive services) for the entire sample period.
- These two service categories together accounted for more than two-thirds of the total commercial services export from India.
- Furthermore, according to an alternative CA measure that considers intra-industry trade, India seems to have CA over the rest of the world in different services such as travel, communication services, and personal, cultural, and related services as well.

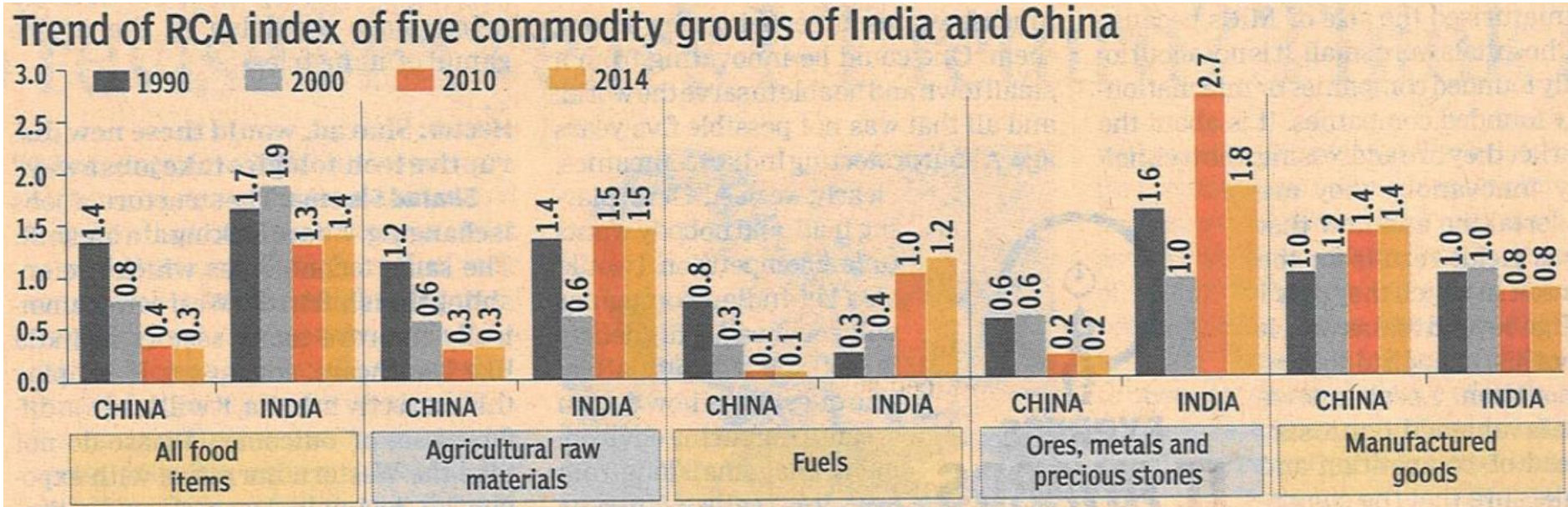
Hiranya K. Nath & Binoy Goswami, 2018. "[India's comparative advantages in services trade](#)," [Eurasian Economic Review](#), Springer; Eurasia Business and Economics Society, vol. 8(2), pages 323-342, August.



India Vrs China - RCA

- the RCA to assess the comparative advantage of the two countries relative to the world average shows that, contrary to widespread belief, there are a number of areas where India has a comparative advantage vis-à-vis China.
- when it comes to commodities like food, agricultural raw materials, ores, metals, precious stones and fuels, Indian exports enjoy a distinct advantage vis-à-vis China.
- It is only in respect of manufactured goods that China has an edge over India.

RCA between China and India



groups during the period 1990-2014.

[The Financial Express](#), December 19, 2016

Fact regarding RCA of India

- India has the highest RCA in the production of rice and spices, with value index values of 17.8 and 10.8 in 2023, respectively.
- China has RCA in many products compared to India, and it has the highest RCA in pottery and fabrics, woven , of man-made fabrics with index values of 44 and 4 in 2023, respectively
- UN trade & development (UNCTAD) provides annual revealed comparative advantage index value of international merchandise trade .
<https://unctadstat.unctad.org/datacentre/dataviewer/US.RCA>

*Thank
you*

